

## Evaluate and Improve Chunking and Ranking, and Understand Graph RAG

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Evaluate the existing Graph RAG retrieval approach using 200 XML articles and compare it with vector search. Build a common dataset for both approaches, evaluate the retrieved top-K chunks for different queries, identify missed or incorrectly ranked chunks, and improve chunking, ranking, and retrieval quality.

Also explore RapidMiner and review the existing Graph RAG implementation to understand the current approach and identify possible improvements.

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### Acceptance Criteria

- **200 XML Articles:** Use the same 200 XML articles for Vector Search and Graph RAG.
- **Top-K Evaluation:** Compare the Top-K chunks returned by both methods.
- **Retrieval Quality:** Check which method gives better and more relevant chunks.
- **Missed Chunks:** Identify relevant chunks that are missed.
- **Chunking:** Understand the current chunking method.
- **Ranking:** Understand how the chunks are ranked.
- **Retrieval:** Understand the current retrieval process.
- **RapidMiner:** Explore RapidMiner and understand how it can help.
- **Graph RAG:** Review the existing Graph RAG flow and make sure it is working correctly.
- **Findings:** Document the results and recommended improvements.

### Tasks under the PBI

#### Task 1 — Evaluate Graph RAG and Vector Search on 200 XML Articles

##### Description:

Create a common dataset of 200 XML articles that can be used by both Graph RAG and Vector Search.

Run the same set of queries against both approaches and compare the top-K retrieved chunks. Check which approach returns more relevant chunks, identify missed chunks, and analyze the differences in retrieval results.

## **Task 2 — Build Vector Search on 200 XML Articles**

### **Description:**

Create and prepare 200 XML articles for Vector Search.

Generate the required chunks and embeddings, store them in the vector search system, and implement the retrieval flow to return the top-K relevant chunks for a given query.

Ensure that the same 200 XML articles are also available for the Graph RAG evaluation.

## **Task 3 — Explore RapidMiner**

### **Description:**

Explore RapidMiner in detail to understand its main capabilities, features, and how it can be used for data processing, search, retrieval, and evaluation.

Identify whether RapidMiner can be useful for the current Graph RAG and Vector Search work and document the relevant findings.

## **Task 4 — Improve Chunking, Ranking, and Retrieval**

### **Description:**

Review the current chunking, ranking, and retrieval approach.

Analyze the retrieved top-K chunks for different queries and identify cases where relevant chunks are missed, irrelevant chunks are returned, or the ranking order is incorrect.

Investigate possible improvements to the chunking strategy, ranking logic, and retrieval process, and test the improvements on the evaluation dataset.

## **Task 5 — Understand Existing Graph RAG**

### **Description:**

Review the existing Graph RAG implementation to understand the complete retrieval flow.

Check how the query is processed, how seed nodes are identified, how the graph is traversed, how related nodes and chunks are retrieved, and how the final top-K chunks are selected and ranked.

Verify that the current implementation is working as expected and document any issues or areas that need improvement.